A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.1: Central Processing Unit (CPU) Architecture

Concept 4.1.2: CPU Components (ALU, CU, Clock, IAS)

Total Questions: 7

Mark Schemes begin on Page 5

Question 1 | Source: 9618_s25_qp_13 (Q2.a.i)

2 A computer has a processor.

(a) The processor has a Control Unit (CU) and system clock.

(i) Explain how the CU and the system clock work together.

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.... [2]

Question 2 | Source: 9618_w24_qp_11 (Q3.a)

3 A computer system has a dual-core Central Processing Unit (CPU).

(a) State the purpose of the system clock and the Control Unit (CU) in a CPU.

System clock ....

....

CU ....

....

[2]

Question 3 | Source: 9618_w23_qp_11 (Q5.b)

5 The Central Processing Unit (CPU) of the basic Von Neumann model for a computer system contains several special purpose registers.

(b) Describe what is meant by the **Immediate Access Store (IAS)** in a computer system.

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.... [2]

Question 4 | Source: 9618_s23_qp_13 (Q7.b)

7 A computer stores data in binary form.

(b) The computer has a Control Unit (CU), system clock and control bus.

Explain how the CU, system clock and control bus operate to transfer data between the components of the computer system.

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.... [4]

Question 5 | Source: 9618_w22_qp_11 (Q5.b.iii)

(b) A Central Processing Unit (CPU) contains several special purpose registers and other components.

(iii) Describe the purpose of the Control Unit (CU) in a CPU.

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.... [2]

Question 6 | Source: 9618_w22_qp_13 (Q4.b)

(b) A computer system contains a system clock.

Describe the purpose of the system clock.

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.... [2]

Question 7 | Source: 9618_s21_qp_12 (Q5.a)

5 Seth uses a computer for work.

(a) Complete the following descriptions of internal components of a computer by writing the missing terms.

The .... transmits the signals to coordinate events based

on the electronic pulses of the ....

The .... carries data to the components, while the

.... carries the address where data needs to be written to

or read from.

The .... performs mathematical operations and

logical comparisons.

[5]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_13 (Q2.a.i)

2(a)(i)	**1 mark** each to **max 2 - max 1** for each set of marks • The control unit synchronises the actions of the processor • by sending a command / signal on each timing signal produced by the system clock • using / along the control bus	**2**

Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q3.a)

3(a)	**1 mark** for system clock and **1 mark** for Control Unit System clock • To synchronise operations • ... by creating and transmitting timing signals on the control bus Control Unit • Sends/receives control signals along control bus • Reads an instruction from the contents of the memory location whose address is stored in PC • Coordinates/synchronises the activity of other components in the CPU • Manages the execution of instructions • Controls communication between the components in the CPU	**2**

Mark Scheme for Question 3 | Source: 9618_w23_ms_11 (Q5.b)

5(b)	**1 mark** for each bullet point (**max 2**) • Immediate Access Store holds all the data / instructions / programs currently in use • Immediate Access Store is volatile memory • Immediate Access Store has fast access times	**2**

Mark Scheme for Question 4 | Source: 9618_s23_ms_13 (Q7.b)

7(b)	**1 mark** each to **max 4** • The **system clock** gives out **timing** signals • … which are sent on the **control bus** • …to **synchronise** the other system components • The **Control Unit** initiates data transfer • …by generating signals that are sent on the **control bus** to other components	**4**

Mark Scheme for Question 5 | Source: 9618_w22_ms_11 (Q5.b.iii)

5(b)(iii)	**1 mark** for each bullet point (**max 2**): • to coordinate / synchronise the actions of other components in the CPU • to send / receive control signals along the control bus • to manage the execution of instructions (in sequence) • to control the communication between the components of the CPU	**2**

Mark Scheme for Question 6 | Source: 9618_w22_ms_13 (Q4.b)

4(b)	**1 mark** for each bullet point (**max 2**): • synchronise operations • ... by creating timing signals • to keep track of the date and time / timestamp files • to process operations in the correct order / sequence	**2**

Mark Scheme for Question 7 | Source: 9618_s21_ms_12 (Q5.a)

5(a)	**1 mark** for each term correctly inserted The **control unit/bus** transmits the signals to coordinate events based on the pulses of the (**system) clock**. The **data bus** carries data to components, while the **address bus** carries the address where data is being written to or read from. The **arithmetic logic unit/ALU** performs mathematical operations and logical comparisons.	**5**